

4.4 Taylor's Theorem & The Convergence of Taylor Series

So far we have Taylor polynomials to approximate curves. In the last section we saw that these Taylor polynomials had intervals on which they “converged” or aligned with the desired function very well. The obvious question to ask from here is: When does a Taylor series of a function converge to the *entire* function?

Definition: Suppose f is differentiable n times at x_0 . The n th-degree Taylor polynomial of f at x_0 is

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k \text{ and } R_n(x) = f(x) - p_n(x) \text{ is called the } \underline{n^{\text{th}} \text{ remainder}} \text{ of the Taylor series.}$$

Theorem: Suppose f is infinitely differentiable at x_0 . If $f(x) = p_n(x) + R_n(x)$ (where $p_n(x)$ is the n th-degree Taylor polynomial of f at x_0) and $\lim_{n \rightarrow \infty} R_n(x) = 0$ for $|x - x_0| < R$, then f is equal to the sum of its Taylor series on the interval $(x_0 - R, x_0 + R)$.

Proof: We note that by the given, $\lim_{n \rightarrow \infty} f(x) = \lim_{n \rightarrow \infty} (p_n(x) + R_n(x))$. So, for $|x - x_0| < R$, we have

$\lim_{n \rightarrow \infty} f(x) = \lim_{n \rightarrow \infty} (p_n(x) + R_n(x)) \Leftrightarrow f(x) = \lim_{n \rightarrow \infty} p_n(x)$. We recognize the expression on the right-hand side of this equation as the Taylor series of f on the interval $(x_0 - R, x_0 + R)$. ■

From the above theorem we see that the key to showing that a function is equal to its Taylor series, is showing that $\lim_{n \rightarrow \infty} R_n(x) = 0$ for appropriate values of x . So, we need a theorem that provides information about $R_n(x)$.

We will call this theorem Taylor's Theorem.

Theorem (Taylor's Theorem): Suppose f is differentiable $n+1$ times at x_0 . If $|f^{(n+1)}(x)| \leq M$ for an interval containing the number x_0 , say $a \leq x \leq b$, then the remainder of $R_n(x)$ of the Taylor series

$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$ satisfies the inequality

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1} \text{ for all } x \text{ in } [a, b]$$

Note: This theorem goes by many different names: Taylor's Theorem, Taylor's inequality, & The Remainder Estimation Theorem.

Note: In this theorem n represents the degree of the Taylor polynomial and not merely the final index value. This is important when working with Taylor series that have indexes that get rewritten such as with $\cos(x)$, $\sin(x)$, $\cosh(x)$, $\sinh(x)$, etc.

Proof: (NOT DONE IN CLASS)

We are assuming that f can be differentiated $n+1$ times on an interval contain the number x_0 , say $a \leq x_0 \leq b$ and that $|f^{(n+1)}(x)| \leq M$ for all x in $[a, b]$.

We want to show that: $|R_n(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$ for all x in $[a, b]$.

We first note a few things:

- (1) $R_n(x) = f(x) - p_n(x) = f(x) - \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$
- (2) $R_n(x_0) = R'_n(x_0) = R''_n(x_0) = \dots = R^{(n)}_n(x_0) = 0$ Since,

$$R_n(x) = f(x) - \left[f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!} (x - x_0)^2 \dots \right]$$

$$R'_n(x) = f'(x) - [0 + f'(x_0) + f''(x_0)(x - x_0) \dots] \Rightarrow R'_n(x_0) = f'(x_0) - [f'(x_0) + f''(x_0)(x_0 - x_0) \dots] = 0$$

$$\dots$$
- (3) $R_n^{(n+1)}(x) = f^{(n+1)}(x)$ for all x in $[a, b]$. Since the $(n+1)^{\text{st}}$ derivative for $p_n(x) = 0$.

We will consider the case where $x_0 < x$ that is $(a \leq x_0 < x \leq b)$, this guarantees that $(x - x_0) > 0$ which is easier than the case where $x_0 > x$. Both cases need to be proven to prove the theorem, however we will only investigate the first case.

So we start with the assumption of the theorem: $|f^{(n+1)}(x)| \leq M$. By (3) we can say $|R_n^{(n+1)}(x)| \leq M$ which implies:

$$-M \leq R_n^{(n+1)}(x) \leq M$$

Therefore,

$$\int_{x_0}^x -M dt \leq \int_{x_0}^x R_n^{(n+1)}(t) dt \leq \int_{x_0}^x M dt \quad (\text{by properties of integrals})$$

We note that $\int_{x_0}^x R_n^{(n+1)}(t) dt = R_n^{(n)}(t) \Big|_{x_0}^x = R_n^{(n)}(x) - R_n^{(n)}(x_0)$, but by (2) $R_n^{(n)}(x_0) = 0$, so $\int_{x_0}^x R_n^{(n+1)}(t) dt = R_n^{(n)}(x)$.

From this fact and the previous string of inequalities we have:

$$\int_{x_0}^x -M dt \leq \int_{x_0}^x R_n^{(n+1)}(t) dt \leq \int_{x_0}^x M dt \quad \text{which implies} \quad -M(x - x_0) \leq R_n^{(n)}(x) \leq M(x - x_0).$$

By integrating again and use (2) we have: $-\frac{M}{2}(x - x_0)^2 \leq R_n^{(n-1)}(x) \leq \frac{M}{2}(x - x_0)^2$.

If we keep integrating, then after $n+1$ integrations we will obtain:

$$-\frac{M}{(n+1)!} (x - x_0)^{n+1} \leq R_n(x) \leq \frac{M}{(n+1)!} (x - x_0)^{n+1}$$

By rewriting this we obtain: $|R_n(x)| \leq \frac{M}{(n+1)!} (x - x_0)^{n+1}$ and since $x > x_0$ we write: $|R_n(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$

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Example 1: Consider the function $f(x) = e^x$.

a) Use an appropriate Taylor polynomial of degree 2 to find a good estimate for e^1 .

We will construct a Taylor polynomial centered at 0 to estimate this value.

$$\text{So we want to find: } p_2 = \frac{f(0)}{0!} + \frac{f'(0)}{1!}(x-0) + \frac{f''(0)}{2!}(x-0)^2$$

We note that $f^{(n)}(0) = 1$ for all values of n .

Therefore we have:

$$f(x) \approx p_2 \quad \text{which mean } e^x \approx 1 + (x-0) + \frac{1}{2!}(x-0)^2 = 1 + x + \frac{1}{2}x^2 \quad \text{for values of } x \text{ near } 0.$$

$$\text{Thus } f(1) = e^1 \approx 1 + (1) + \frac{1}{2}(1)^2 = 1 + 1 + \frac{1}{2} = 2.5$$

b) Use Taylor's Theorem to find an upper bound on the error of this approximation.

We note that $f(x) = e^x$ is infinitely differentiable at all points and that on the interval $[0, 1]$,

$|f^{(n+1)}(x)| \leq e^1 = e < 3 = M$. Therefore, we have satisfied the conditions to use Taylor's Theorem.

Taylor's Theorem states that $|R_n(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$ for all x in $[a, b]$. Note that we have the following

values based on our work in part (a) and part (b): $x = 1$, $x_0 = 0$, $n = 2$, $M = 3$, $a = 0$, & $b = 1$.

Therefore, we have,

$$|R_2(1)| \leq \frac{3}{(2+1)!} |1-0|^{2+1} = \frac{3}{6} = \frac{1}{2} \text{ for all } x \text{ in } [0, 1].$$

Therefore, we know that the error of the approximation $e^1 \approx 2.5$ is no larger than $\frac{1}{2}$.

Example 2: Use Taylor's Inequality to determine a value of n you could use to approximate e accurately to two decimal places.

We know that $e = \sum_{n=0}^{\infty} \frac{1}{n!}$. If we want to estimate 2 decimal places accurately then we want to choose n so that the absolute value of the n th remainder at $x=1$ satisfies: $|R_n(1)| < 0.005$. By Taylor's Inequality we know that:

$$|R_n(1)| \leq \frac{e}{(n+1)!} < \frac{3}{(n+1)!}$$

Therefore, we want:

$$\frac{3}{(n+1)!} < 0.005 \quad \Leftrightarrow \quad (n+1)! > 600 \quad \Leftrightarrow \quad n \geq 5 \quad \text{since } (5+1)! = 6! = 720.$$

So we obtain an *exact* value of e to two decimal places with: $e \approx 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} \approx 2.71$

Note: You could easily imagine that a computer coded to compute up to millions of places of accuracy very quickly using the power of Taylor's Theorem.

Example 3: Prove that $f(x) = e^x$ is equal to its Maclaurin series for all x .

We want to show that $\lim_{n \rightarrow \infty} R_n(x) = 0$ for all x . So, let x be an arbitrary real number. We know that there must exist some positive number d so that $|x| \leq d$.

Now applying Taylor's Inequality. We note that if $|x| \leq d$, then $|f^{(n+1)}(x)| = |e^x| \leq e^d$ and this holds for every value of n . Now, by Taylor's inequality we have $0 \leq |R_n(x)| \leq \frac{e^d}{(n+1)!} |x|^{n+1}$ for $|x| \leq d$ and each value of n .

Now, we note that $\lim_{n \rightarrow \infty} \frac{e^d}{(n+1)!} |x|^{n+1} = e^d \lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{(n+1)!} = 0$ (see 5.1.2 notes for details). So, by the squeeze theorem we have that $\lim_{n \rightarrow \infty} |R_n(x)| = 0$ for $|x| \leq d$ which immediately implies that $\lim_{n \rightarrow \infty} R_n(x) = 0$ for $|x| \leq d$.

Thus, $f(x) = e^x$ is equal to its Maclaurin series for all x . That is, $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for all values of x .

Example 4: Assuming $x = 1$, what does example 1 tell us must be true?

This means that $e = \sum_{n=0}^{\infty} \frac{1}{n!}$. Therefore we can obtain the exact decimal places of e by writing out terms of this polynomial for long enough!!!

Note: Using a partial sum of this series will only allow us to estimate the value of e and we can approximate the error of each approximation using Taylor's Theorem. However, the infinite series produces the exact value of e !

Example 5: Consider the function $f(x) = \sin(x)$.

- a) Use the Maclaurin series for $f(x) = \sin(x)$, $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$, to create a 7th degree polynomial to estimate the value of $\sin(1)$.

If $n = 3$ we obtain a 7th degree polynomial. Therefore we have:

$$f(x) \approx p_3 \quad \text{which mean } \sin(x) \approx x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} \quad \text{for values of } x \text{ near } 0.$$

$$\text{Thus } f(1) = \sin(1) \approx (1) - \frac{(1)^3}{6} + \frac{(1)^5}{120} - \frac{(1)^7}{5040} = \frac{4241}{5040} \approx 0.8415$$

- b) Use Taylor's Theorem to find an upper bound on the error of this approximation.

We note that $f(x) = \sin(x)$ is infinitely differentiable at all points and that $f^{(n+1)}(x)$ is $\pm \cos(x)$ or $\pm \sin(x)$. This means that $|f^{(n+1)}(x)| \leq 1 = M$ for all values of x . Therefore, we have satisfied the conditions to use Taylor's Theorem.

Note that we have the following values based on our work in part (a) and part (b):

$x = 1$, $x_0 = 0$, $n = 7$, $M = 1$, $a = -\infty$, & $b = \infty$. Therefore we have,

$$|R_7(1)| \leq \frac{1}{(7+1)!} |1-0|^{7+1} = \frac{1}{40320} \text{ for all } x \text{ in } (-\infty, \infty).$$

Therefore we know that the error of the approximation $\sin(1) \approx 0.8415$ is no larger than $\frac{1}{40320}$.

Example 6: Use Taylor's Inequality to determine a value of n you could use to approximate $\sin(1)$ accurately to 10 decimal places.

If we want to estimate 10 decimal places accurately then we want to choose n so that the absolute value of the n th remainder at $x = 1$ satisfies: $|R_n(1)| < 0.00000000005$. We note again that $f^{(n+1)}(x)$ is $\pm \cos(x)$ or $\pm \sin(x)$. This means that $|f^{(n+1)}(x)| \leq 1$ and this holds for every value of n and x . So, by Taylor's Inequality we know that:

$$|R_n(1)| \leq \frac{1}{(n+1)!} (1)^{n+1} = \frac{1}{(n+1)!}$$

Therefore, we want: $\frac{1}{(n+1)!} < 0.00000000005 \Leftrightarrow (n+1)! > 20,000,000,000 \Leftrightarrow n \geq 13$ since

$$(13+1)! = 14! = 87,178,291,200.$$

So we obtain an *exact* value of $\sin(1)$ to 10 decimal places with:

$$\sin(1) \approx 1 - \frac{1}{3!} + \frac{1}{5!} - \frac{1}{7!} + \frac{1}{9!} - \frac{1}{11!} + \frac{1}{13!} - \frac{1}{15!} + \frac{1}{17!} - \frac{1}{19!} + \frac{1}{21!} - \frac{1}{23!} + \frac{1}{25!} \approx 0.8414709848$$

Example 7: Prove that $f(x) = \sin(x)$ is equal to its Maclaurin series for all x .

Let x be an arbitrary real number. We know that there must exist some positive number d so that $|x| \leq d$. We need to show that if $R_n(x)$ is the remainder of the Maclaurin series for f then $\lim_{n \rightarrow \infty} R_n(x) = 0$ for $|x| \leq d$.

To do this, we will use Taylor's Inequality. We note that $f^{(n+1)}(x)$ is $\pm \cos(x)$ or $\pm \sin(x)$. This means that $|f^{(n+1)}(x)| \leq 1$ and this holds for every value of n . Now, by Taylor's inequality we have

$$0 \leq |R_n(x)| \leq \frac{1}{(n+1)!} |x|^{n+1} \text{ for } |x| \leq d \text{ and each value of } n.$$

Now, we note that $\lim_{n \rightarrow \infty} \frac{1}{(n+1)!} |x|^{n+1} = \lim_{n \rightarrow \infty} \frac{|x|^{n+1}}{(n+1)!} = 0$ (see 3.1.2 notes for details). So, by the squeeze theorem

we have that $\lim_{n \rightarrow \infty} |R_n(x)| = 0$ for $|x| \leq d$ which immediately implies that $\lim_{n \rightarrow \infty} R_n(x) = 0$ for $|x| \leq d$.

Thus, $f(x) = \sin(x)$ is equal to its Maclaurin series for all x .

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